Thus, the total cost is

$$\boxed{\text{Total Cost} = \frac{n}{h} \cdot c \cdot (W \cdot p_{\text{in}} + T \cdot p_{\text{out}})\,.}$$

For a uniform hash where $c \approx 2^h$, this can be approximated as

$$\text{Total Cost} \approx \frac{n}{h} \cdot 2^h \cdot (W \cdot p_{\text{in}} + T \cdot p_{\text{out}})\,.$$

We empirically measure the cost of hiding a byte of data in USD by measuring the difference in API costs on the OpenAI usage dashboard before and after each trial. We hide 1 byte of data for both the random projection and PCA-based LSH. Additionally, we set a cutoff of 10 cents per attempt, as the system may enter a low-entropy state where the next continuations are overwhelmingly likely to all hash to a single output. When this happens, we choose to sample an incorrect choice and have error correction fix this on recovery. Table 1 contains the results. We observe that PCA has an overall lower cost when entropy is high enough, but low-entropy situations cause the total cost to be higher. Given that the PCA LSH has a higher accuracy than the Random LSH (Section 7.3), the optimal choice depends several factors including, length of message, covertext distribution, bandwidth, financial and covertext budgets, and choice in error correcting code.

LSH Scheme	**Success Rate (%)**	**Avg Cost (Success) (¢/Byte)**	**Avg Cost (Total) (¢/Byte)**
Random	100%	0.7	0.7
PCA	95%	0.35	0.85

Table 1: Success rate and cost comparison of different LSH schemes for our steganography system taken over 20 trials of hiding 1 byte. Due to low-entropy situations, the next continuation may overwhelmingly be likely to hash to the same value. We therefore cap the cost of hiding a byte at 10¢. The system then sample the wrong value and relies on error correction during retrieval. "Avg Cost (Success)" is the average cost per successfully hidden byte, ignoring failures. "Avg Cost (Total)" includes failures at 10¢ per attempt in the cost.

Comparison of the Two Systems We provide a direct comparison between the watermarking-based and embedding-based approaches, focusing on covertext overhead and resistance to attacks.

Covertext Overhead: Plotting hidden message length vs. covertext length to illustrate efficiency differences. Figure 7 compares the Watermarking and Embedding system for various parameters. The embedding system uses substantially less tokens than the watermarking system for all configurations. This includes configurations of the Watermarking system with high levels of delta ($\delta \geq 0.1$) which can significantly degrades text quality.

Resistance to Attack: Contrasting message recovery under varying levels of n-gram shuffling, synonym substitutions, and AI paraphrasing. Figure 8 contains the results of attacking various texts generated via the watermarking system and Figure 9 contains the results of attacking various texts generated via the embedding system. For the watermarking system, attacks were run in a local mode that contained modifications to individual sentences and a global mode that attacked

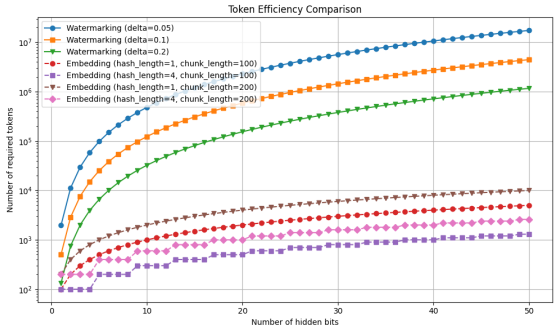


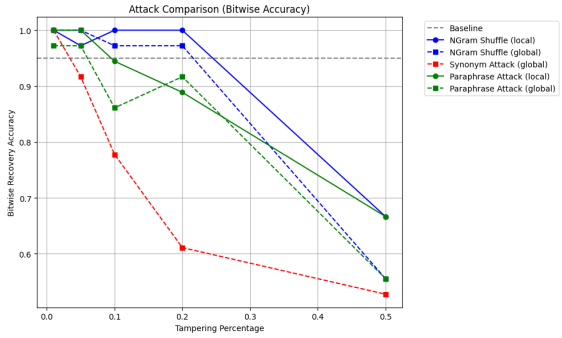
Fig. 7: Covertext Overhead. The x-axis contains the number of bits being hidden within the covertext and the y-axis contains the required length of the covertext measured in tokens to hide those bits. The watermarking schemes require orders of magnitude more tokens to hide the same length message as the embedding scheme, even for high levels of delta that would not be used in practice.

the entire text at once. These attacks are in some way an upper bound and can be viewed as overestimating an adversaries capabilities. For example, synonyms are chosen via nltk's wordnet and can be inappropriate, replacing words with substitution that change tense or no longer have the same meaning (i.e. replacing not with non). n-gram shuffles can create completely nonsensical sentences (i.e. changing "The white cat jumped on the broken couch." to "the broken couch cat jumped on The white."). This becomes particularly extreme in global mode, where n-grams at the end of long texts can be shuffled together with n-grams at the beginning of a text. Therefore, local attacks may be more realistic in this sense. Additionally, local attacks serve as a more direct comparison between the watermarking and embedding systems. This is due to the challenge from their different threat models and the key difference that hidden bits are "spread out" throughout the entire text in the watermarking case compared to hidden bits being "clumped together" in discrete sections of the text in the embedding case. These different strengths and weaknesses make local attacks on watermarked text a more appropriate point of comparison to attacks on segments of the embedded text.

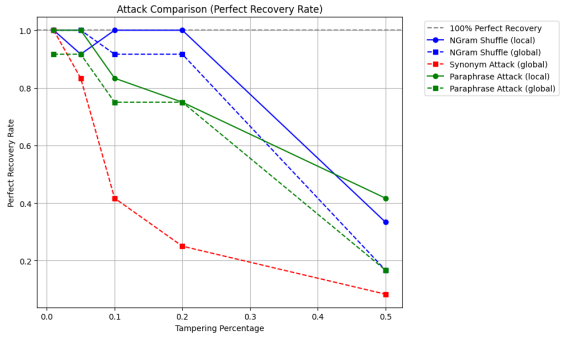
The watermarking system withstands n-gram shuffling attacks well up to 20% of the text being tampered. Local tampering retains a perfect recovery, while global tampering drops to a 90% recovery rate. It handles synonym and paraphrasing attacks up to 5% tampering before quickly declining in both the local and global cases.

The embedding system withstands n-gram shuffling and paraphrasing attacks perfectly at all tested level ranging from 1% to 50% but declines slightly for synonym attacks down to 95% recovery when half of the message blocks are attacked. This is partially from non-perfect agreement in LSH accuracies and some synonym substitutions being inapropriate and changing the meaning of text segments.

Overall, the embedding system substantially outperforms the watermarking system, both in the covertext overhead and its resistance to attacks.



(a) We measure the percentage of bits that are correctly recovered. For example, if [1, 0, 1] was hidden and [0, 0, 1] was recovered, this would contribute a score of 0.67 for that message + attack data point.



(b) We measure whether the bits that are recovered perfectly match the hidden message. For example, if [1, 0, 1] was hidden and [0, 0, 1] was recovered, this would contribute a score of 0 for that message + attack data point.

Fig. 8: We run n-gram shuffle, synonym, and paraphrasing attacks against 12 stegotexts containing 3 bit messages created via the watermarking scheme for various modes and portions of the text being altered.

7.6 Summary of Findings

Our evaluation demonstrates that the embedding system significantly outperforms the watermarking system in both efficiency and robustness to adversarial attacks, while operating at a reasonable cost.

Covertext Efficiency: The embedding-based approach requires substantially fewer tokens to encode the same number of hidden bits compared to the watermarking-based approach. Even with high perturbation levels ($\delta \geq 0.1$) that degrade text quality, the watermarking system remains significantly less efficient.

Resistance to Attacks: The watermarking system maintains strong recovery under minor modifications (e.g., $\leq$20% n-gram shuffling) but quickly degrades when subjected to synonym substitutions or paraphrasing attacks. The embedding system demonstrates near-perfect robustness to n-gram shuffling and paraphrasing attacks. However, synonym substitution causes slight degradation (95% recovery at 50% attack rate), though this is partly due to inappropriate synonym choices that lay outside of our threat model rather than inherent system weaknesses.

Rejection Sampling and Cost: The PCA LSH achieves higher accuracy than the Random Projection LSH (91.6% vs. 87.6% in first-message sampling tests). Rejection sampling for the embedding system requires fewer iterations for the PCA LSH (2.03 samples on average) compared to the Random Projection LSH (2.82 samples). The PCA LSH achieves lower cost per successful hidden byte (0.35¢) but is more affected by low-entropy situations, leading to higher total cost when failures are included (0.85¢/byte vs. 0.7¢/byte for the random projection).

Practical Considerations: The embedding system is preferred when strong robustness is required against adversarial paraphrasing. The watermarking system, despite being more fragile, might still be useful in scenarios where minor modifications are expected, and message length constraints are less strict. One key advantage of the watermarking system is that it hides hidden bits in a manner that is diffused throughout the entire text, where no particular section is tied to a particular bit. The embedding system lacks this feature.